

4.6 Waves

AQA GCSE Physics (8463) — Paper 2

4.6.1 Transverse and Longitudinal Waves

A **wave** is a repeating oscillation (vibration) that transfers energy from one place to another without permanently moving the matter itself. Only the energy travels; particles oscillate about fixed positions and return to equilibrium.

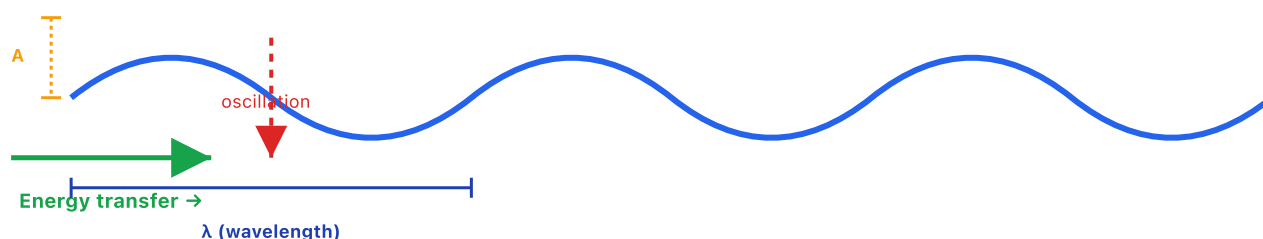
Transverse Waves

In a **transverse wave**, particles of the medium oscillate **perpendicular (at right angles) to the direction of energy transfer**. As the wave moves forward, particles move up and down (or side to side).

Examples of transverse waves:

- Water ripples and ocean waves (water particles bob up and down)
- Waves on a rope or string (particles move side to side)
- **All electromagnetic (EM) waves:** light, radio waves, X-rays, etc. (electric and magnetic fields oscillate perpendicular to direction of travel)

Transverse Wave (e.g. rope wave)

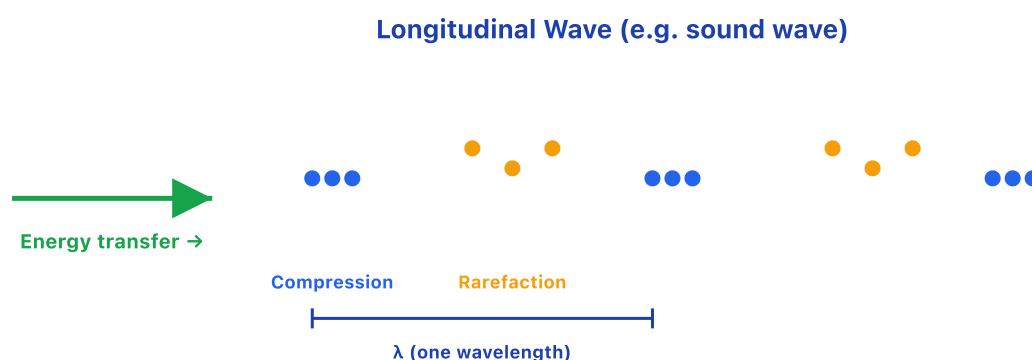


Longitudinal Waves

In a **longitudinal wave**, particles of the medium oscillate **parallel (in the same direction) to the direction of energy transfer**. The wave consists of alternating regions of compression (particles pushed together) and rarefaction (particles spread apart).

Examples of longitudinal waves:

- Sound waves in air, water, and solids (compressions and rarefactions of air molecules)
- P-waves (primary seismic waves) from earthquakes (compressions of rock layers)



Exam Tip: The key difference: transverse = particles move PERPENDICULAR to wave travel; longitudinal = particles move PARALLEL to wave travel. Remember: light and string waves are transverse; sound waves are longitudinal. These cannot be confused in exam questions.

4.6.2 Properties of Waves

Amplitude, Wavelength, Frequency, and Period

Amplitude (A) is the maximum displacement of a particle from its equilibrium position. It is the height of a peak (or depth of a trough) from the centre line. Measured in metres (m).

Wavelength (λ , lambda) is the distance from one point on a wave to the next equivalent point — typically measured from peak to peak, or trough to trough, or compression to

compression. Measured in metres (m).

Frequency (f) is the number of complete waves (oscillations) produced in one second. Measured in hertz (Hz). $1 \text{ Hz} = 1 \text{ wave per second}$.

Period (T) is the time taken for one complete oscillation or one complete wave to pass a point. Measured in seconds (s). Relationship: $T = 1 / f$

The Wave Equation

The **wave speed (v)** is related to frequency and wavelength by:

$$v = f \times \lambda$$

where **v** is wave speed (m/s), **f** is frequency (Hz), and **λ** is wavelength (m).

Worked Examples

Example 1: A wave on a string has a wavelength of 2 m and a frequency of 50 Hz. Calculate the wave speed.

$$v = f \times \lambda = 50 \times 2 = 100 \text{ m/s}$$

Example 2: A sound wave in air travels at 340 m/s with a frequency of 680 Hz. Calculate the wavelength.

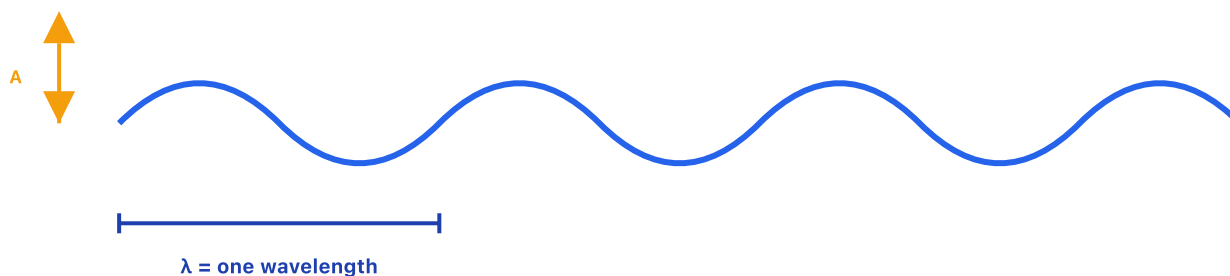
$$\begin{aligned} v &= f \times \lambda \\ 340 &= 680 \times \lambda \\ \lambda &= 340 / 680 = 0.5 \text{ m} \end{aligned}$$

Example 3: A ripple tank generates waves with a period of 0.2 s and wavelength of 4 m. Find the wave speed.

First find frequency: $f = 1 / T = 1 / 0.2 = 5 \text{ Hz}$

Then: $v = f \times \lambda = 5 \times 4 = 20 \text{ m/s}$

Wave Properties



Period T = time for one complete wave to pass = time from one peak to the next peak

Exam Tip: In calculations, always ensure units match: if wavelength is in metres, frequency in Hz, then speed comes out in m/s. If you're given wavelength in cm, convert to m first. Period T and frequency f are reciprocals: $T = 1/f$ and $f = 1/T$ — never mix them up.

Required Practical 8: Measuring Wave Speed

Students measure the frequency, wavelength, and speed of waves using:

- **String/rope vibration:** Use a vibration generator and measure wavelength with a ruler; frequency from the generator
- **Ripple tank:** Create waves in shallow water, measure wavelength using a ruler or grid, frequency from the vibrator
- **Ultrasound pulses:** Use ultrasound transmitter and receiver to measure distance and time of travel

4.6.3 Sound Waves (Triple)

Sound is a longitudinal wave produced by vibrating objects. Sound waves are compressions and rarefactions of the medium (air, water, solids) and travel outward from the source.

The Range of Human Hearing

The human ear can detect sound frequencies from approximately **20 Hz to 20 000 Hz (20 kHz)**. Sound outside this range is inaudible to humans:

- **Infrasound:** frequencies below 20 Hz (elephant calls, earthquakes)
- **Audible range:** 20 Hz to 20 kHz
- **Ultrasound:** frequencies above 20 kHz (bat echolocation, medical imaging)

Pitch and Loudness

Pitch is determined by frequency:

- Higher frequency → higher pitch (treble, squeaky sounds)
- Lower frequency → lower pitch (bass, deep sounds)

Loudness is determined by amplitude:

- Larger amplitude → louder sound (more energy)
- Smaller amplitude → quieter sound (less energy)

The Human Ear

Sound waves enter the ear and set the **eardrum (tympanum)** vibrating. The vibrations are transferred through three tiny bones called **ossicles** (hammer, anvil, stirrup) to the **cochlea** — a fluid-filled spiral structure. The cochlea contains hair cells that convert mechanical vibrations into electrical nerve signals, which travel to the brain for perception.

Speed of Sound in Different Media

Medium	Speed of Sound	Why?
Solids (e.g. steel, wood)	~5000 m/s	Particles tightly packed; vibrations transfer quickly through strong bonds
Liquids (e.g. water)	~1500 m/s	Particles more loosely packed than solids but closer than gases

Gases (air at 20°C)	~340 m/s	Particles far apart; vibrations travel more slowly through empty space
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Exam Tip: Sound travels FASTEST in solids and SLOWEST in gases. This is the opposite of light — light travels fastest in vacuum and slows in denser media. In exam questions, never say sound travels at "the speed of sound" without specifying the medium.

4.6.4 Ultrasound (Triple)

Ultrasound is sound with a frequency greater than 20 000 Hz (20 kHz) — beyond the upper limit of human hearing. Ultrasound can be produced by transducers and has many practical applications.

Partial Reflection at Boundaries

When an ultrasound wave encounters a boundary between two media with **different densities** (or different acoustic impedances), part of the wave is reflected back and part is transmitted through. The amount of reflection depends on the density difference between the media. This property is exploited in medical and industrial applications.

Time of Flight and Distance Calculation

Ultrasound pulses are sent into a medium; the echo (reflected pulse) returns to a receiver. By measuring the **time of flight** (total time for the pulse to travel down and back), the distance can be calculated:

Distance to object (one way):

$$s = v \times t$$

where **s** is distance (m), **v** is wave speed (m/s), and **t** is time.

If measuring round-trip time, divide by 2: $\text{distance} = (v \times t) / 2$

Worked Example: Ultrasound Depth Measurement

An ultrasound pulse is sent into water. The echo returns to the transducer after 0.01 s. The speed of ultrasound in water is 1500 m/s. Calculate the depth of the object (distance to the reflecting surface).

$$\begin{aligned}\text{Total distance (down and back)} &= v \times t = 1500 \times 0.01 = 15 \text{ m} \\ \text{Distance to object (one way)} &= 15 / 2 = 7.5 \text{ m}\end{aligned}$$

Medical Applications (Foetal Imaging)

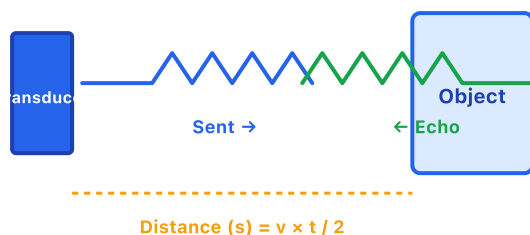
Ultrasound scanning in pregnancy:

- A transducer sends short pulses of ultrasound into the womb
- Echoes reflect off the foetus and surrounding tissues
- A computer calculates distances and builds a real-time image
- No ionising radiation → safer than X-rays (no risk of cell damage or mutation)
- Can measure foetal size, detect abnormalities, confirm due date

Other Applications

- **Industrial flaw detection:** Detect cracks or voids in metal structures, pipes, welds
- **Sonar/echo location:** Submarines and ships map the seafloor; bats locate prey in flight
- **Cleaning:** Ultrasonic cleaning tanks shake jewellery, dentures, and precision instruments to remove dirt
- **Therapeutic ultrasound:** Physiotherapy and pain relief

Ultrasound Pulse and Echo



Sent at $t=0$, echo received at $t = \text{total time}$

Exam Tip: Always remember to divide by 2 when calculating distance from round-trip time — the pulse travels down to the object and back up. A common mistake is forgetting this division, doubling your answer incorrectly.

4.6.5 Seismic Waves (Triple)

Seismic waves are vibrations generated by earthquakes deep inside the Earth. They travel outward through rock layers and are detected by seismographs around the world. There are two main types: P-waves and S-waves.

P-waves (Primary Waves)

- **Type:** Longitudinal (compressions and rarefactions)
- **Speed:** Faster (~6 km/s in the crust)
- **Travel through:** Both solids AND liquids
- **Arrival:** Arrive first at distant seismic stations
- **Nature:** Push-pull motion, less destructive

S-waves (Secondary Waves)

- **Type:** Transverse (up-down, side-to-side motion)
- **Speed:** Slower (~3.5 km/s in the crust)

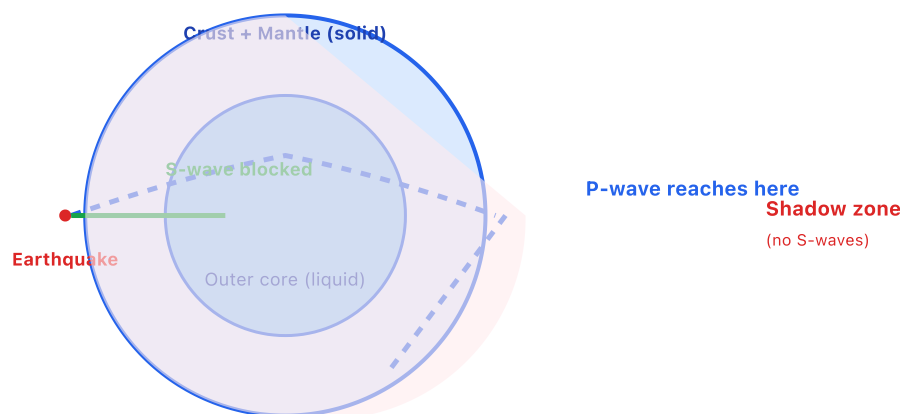
- **Travel through:** Solids ONLY — cannot travel through liquids
- **Arrival:** Arrive second at distant seismic stations
- **Nature:** Shearing motion, more destructive

Evidence for Earth's Structure

The pattern of P and S wave arrivals around the Earth provides evidence for Earth's internal structure:

- **P-waves reach the other side of Earth:** They travel through the liquid outer core
- **S-waves never reach the other side of Earth:** They cannot travel through liquids, so they are absorbed by the outer core. A "shadow zone" exists on the opposite side where no S-waves are detected.
- **This proves the outer core is liquid:** If the outer core were solid, S-waves would pass through it

P-waves and S-waves through Earth



Exam Tip: In exam questions about seismic waves, remember: P = Primary (fast, first to arrive, through liquids), S = Secondary (slow, later arrival, solid only, no liquid). The shadow zone is a key piece of evidence for the liquid outer core — S-waves cannot reach the opposite side because they cannot travel through liquid.

4.6.6 Electromagnetic Spectrum

The **electromagnetic (EM) spectrum** is the entire range of electromagnetic waves, ordered by wavelength or frequency. All EM waves are transverse, are produced by oscillating electric and magnetic fields, and travel at the speed of light in vacuum ($3 \times 10^8 \text{ m/s}$).

Order of the EM Spectrum (Long to Short Wavelength)

From longest wavelength to shortest (or lowest to highest frequency):

1. **Radio waves** — lowest frequency, longest wavelength
2. **Microwaves**
3. **Infrared (heat) radiation**
4. **Visible light** (narrow band: red → orange → yellow → green → blue → indigo → violet)
5. **Ultraviolet (UV)**
6. **X-rays**
7. **Gamma (γ) rays** — highest frequency, shortest wavelength

Key Property: Constant Speed

All EM waves travel at the same speed in vacuum: $c = 3 \times 10^8 \text{ m/s}$. Since $v = f \times \lambda$, waves with higher frequency have shorter wavelength, and vice versa.

Uses of the EM Spectrum

Type	Wavelength (approx)	Uses
Radio waves	1 m – 1 km (and longer)	TV broadcasting, FM/AM radio, WiFi communication
Microwaves	1 mm – 1 m	Satellite communication (TV, phones), microwave ovens (heat water molecules), radar

Infrared (IR)	700 nm – 1 mm	Heating (radiant heaters), remote control signals, thermal imaging cameras, optical fibre communication
Visible light	380 – 700 nm	Human vision, photography, displays, illumination
Ultraviolet (UV)	10 – 380 nm	Fluorescent lamps, sunbeds, sterilisation (kill bacteria), security marks (invisible inks), tanning
X-rays	0.01 – 10 nm	Medical imaging (X-ray photographs of bones/lungs), airport security scanners, industrial inspection
Gamma rays	< 0.01 nm	Cancer treatment (radiotherapy), sterilising medical equipment, medical tracers, nuclear research

Dangers of EM Radiation (Higher Frequency = More Dangerous)

Ionising radiation (UV, X-rays, gamma rays) has enough energy per photon to remove electrons from atoms, creating ions. This causes DNA damage, mutations, and cellular death. Longer-wavelength radiation (radio, microwave, IR) is non-ionising — it transfers energy as heat but does not create ions.

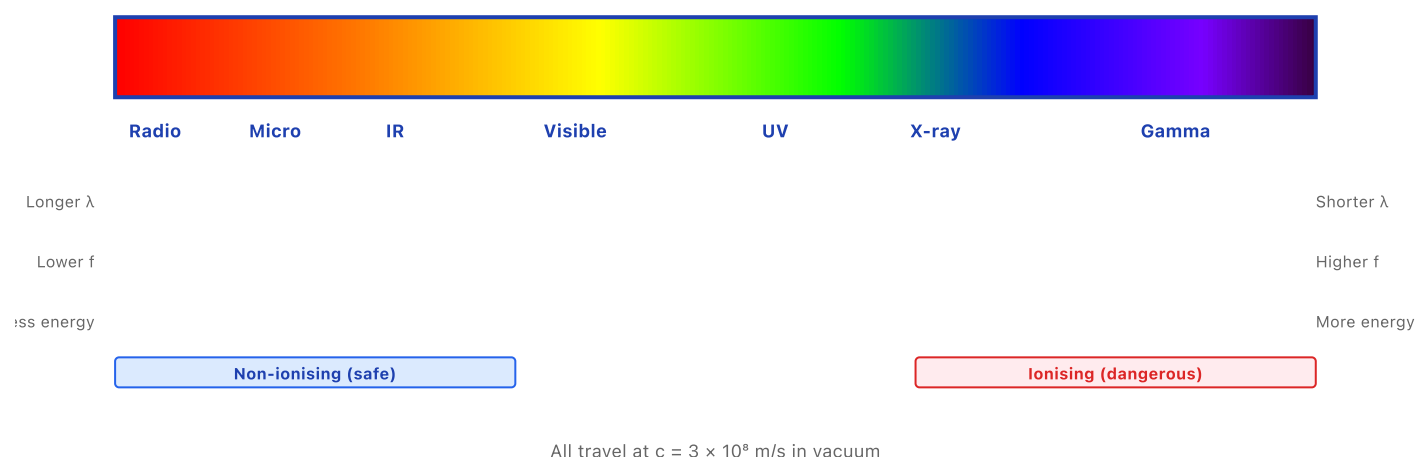
Type	Danger
Ultraviolet	Skin cancer and melanoma (damage to skin cells), cataracts (eye damage), immune system suppression
X-rays and Gamma rays	Ionising radiation causes DNA damage → mutations → cancer. Acute exposure causes radiation sickness. Can damage bone marrow and reproductive organs.

Radiation Dose and Background Radiation

Radiation dose is measured in **sieverts (Sv)**. One sievert is a large dose; most measurements are in millisieverts (mSv) or microsieverts (μSv).

- **Background radiation:** ~2.5 mSv per year on average (natural background from cosmic rays, radon gas, radioactive minerals in Earth)
- **Safe limit:** Occupational exposure ≤ 20 mSv/year; public exposure ≤ 1 mSv/year
- **Single high dose:** 6 Sv or more is likely fatal

Electromagnetic Spectrum



Exam Tip: Higher frequency EM waves are more dangerous because each photon carries more energy. Frequency and wavelength are inversely related: as frequency increases, wavelength decreases. The visible light spectrum is only a tiny portion of the entire EM spectrum — most EM radiation is invisible to the human eye.

4.6.7 Reflection and Refraction

Reflection

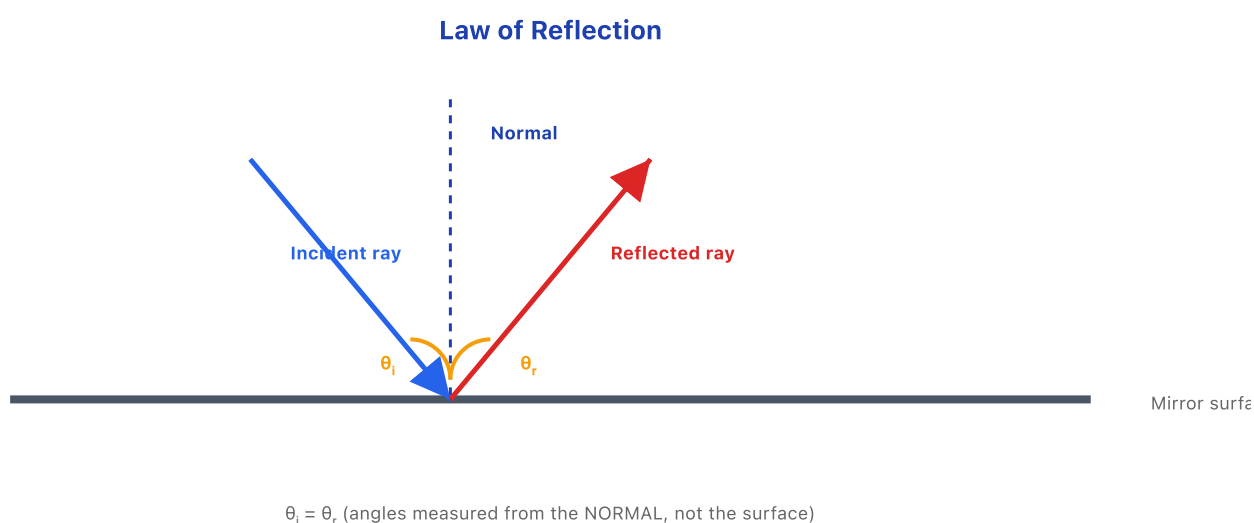
Reflection is when a wave bounces off a surface and returns. The angle of incidence equals the angle of reflection, both measured from the **normal** (an imaginary perpendicular line to the surface).

Law of Reflection: Angle of incidence = Angle of reflection

Types of Reflection

Specular reflection (smooth surface): A smooth, mirror-like surface reflects all parallel incident rays in the same direction. The reflected rays remain parallel. Example: a mirror or still water surface.

Diffuse reflection (rough surface): A rough, bumpy surface reflects incident rays in all different directions (each ray obeys the law of reflection, but normal varies). Parallel light becomes scattered. Example: paper, rough walls, skin. This is how we see non-shiny objects.



Refraction

Refraction is the bending of a wave as it enters a different medium. The wave changes direction because its speed changes in the new medium. Frequency stays the same, but wavelength and speed change. Hence: $v = f \times \lambda \rightarrow$ if v decreases, λ must decrease (for constant f).

Rules for Refraction

Entering a denser medium (slower): e.g. air $\rightarrow$ glass, air $\rightarrow$ water

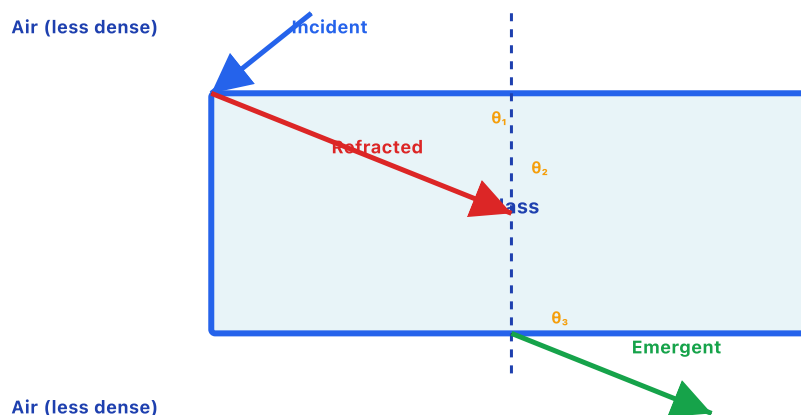
- Wave speed DECREASES
- Wavelength DECREASES

- Wave bends TOWARDS the normal

Entering a less dense medium (faster): e.g. glass → air, water → air

- Wave speed INCREASES
- Wavelength INCREASES
- Wave bends AWAY FROM the normal

Refraction: Ray Diagram through Glass Block

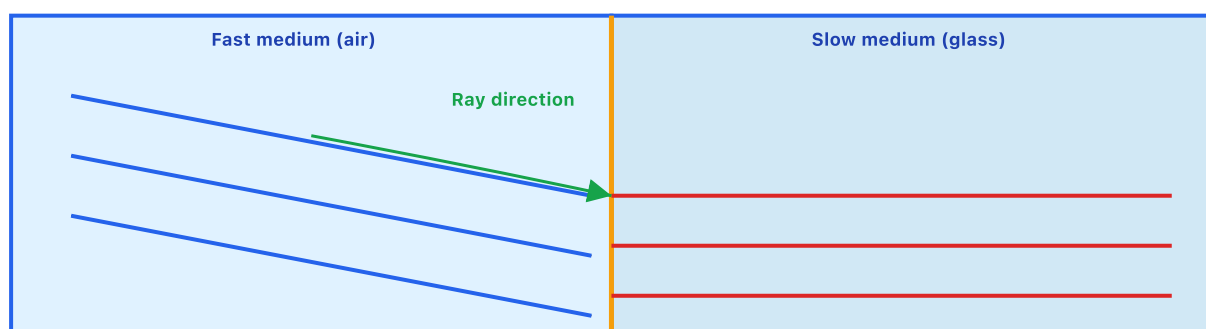


Light bends towards normal (denser) then away (less dense). Emergent ray is parallel to incident ray.

Wavefronts and Refraction

A wavefront is a line (or surface) perpendicular to the direction of wave travel, connecting all points oscillating in phase. When a wavefront crosses a boundary at an angle, one side enters the new medium first and slows down, causing the wavefront to bend. This geometric picture explains why rays also bend.

Wavefronts Bending at a Boundary



Wavefronts space closer in slower medium. Wave bends towards normal (less distance per time = reduced speed).

Required Practical 9: Reflection and Refraction

Students use ray boxes to observe:

- Reflection from plane mirrors (verify law of reflection with angles)
- Refraction through glass blocks (observe bending of light at boundaries)
- Refraction through prisms (light bends twice)
- Total internal reflection (when light travels from dense to less dense medium at steep angle, all light reflects back)

Exam Tip: In refraction, the normal is perpendicular to the surface, NOT parallel to it. Angles of incidence and refraction are measured from the normal. A common mistake is measuring from the surface — this gives wrong angles. Always draw the normal as a dashed perpendicular line.

4.6.8 Lenses (Triple)

A **lens** is a curved, transparent optical element that uses refraction to bend light rays. Lenses are used in cameras, projectors, microscopes, and the human eye to focus or spread light.

Converging Lens (Convex)

A **converging lens** is thicker in the middle and thinner at the edges. It bends parallel light rays towards the optical axis, and all rays meet at a point called the **focal point (F)**.

Focal length (f) is the distance from the centre of the lens to the focal point. For a converging lens, the focal length is the distance at which parallel rays converge. Measured in metres (m) or centimetres (cm).

Diverging Lens (Concave)

A **diverging lens** is thinner in the middle and thicker at the edges. It bends parallel light rays away from the optical axis. The rays appear to come from a virtual focal point behind the lens.

Real and Virtual Images

A **real image** is formed when light rays actually converge at a point. It can be projected onto a screen. Real images are formed by converging lenses when the object is placed beyond the focal length.

A **virtual image** is formed when light rays appear to come from a point, but do not actually meet there. It cannot be projected onto a screen. Virtual images are formed by diverging lenses and by converging lenses when the object is inside the focal length (magnifying glass effect).

Magnification

Magnification (M) is the ratio of image height to object height:

$$M = h_{\text{image}} / h_{\text{object}} = v / u$$

where h is height, v is image distance, and u is object distance (all in same units). If $M > 1$, image is enlarged; if $M < 1$, image is diminished. Negative M means inverted image.

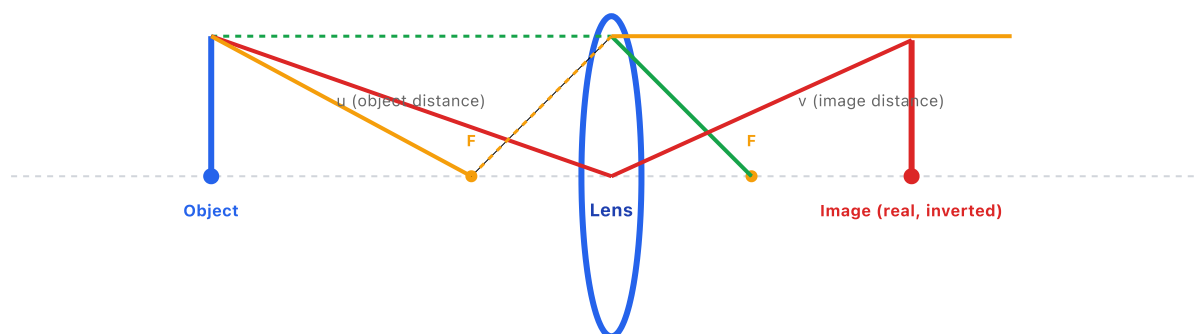
Ray Diagram Rules for Converging Lens

To draw a ray diagram for a converging lens, use three standard rays from the top of the object:

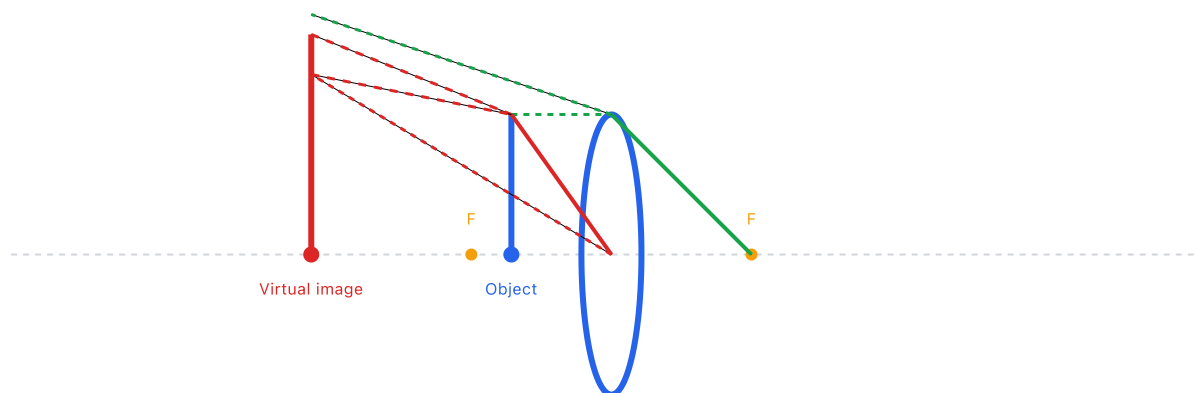
1. **Ray 1 (parallel):** Draw a ray parallel to the optical axis from the top of the object. It refracts through the lens and passes through the focal point F on the far side.
2. **Ray 2 (central):** Draw a ray from the top of the object straight through the centre of the lens. It goes straight through (no bending at the lens centre).
3. **Ray 3 (focal):** Draw a ray from the top of the object through the focal point F on the near side. It refracts through the lens and emerges parallel to the optical axis.

The image is located where rays 1, 2, and 3 intersect on the far side of the lens.

Converging Lens: Real Image (object beyond f)

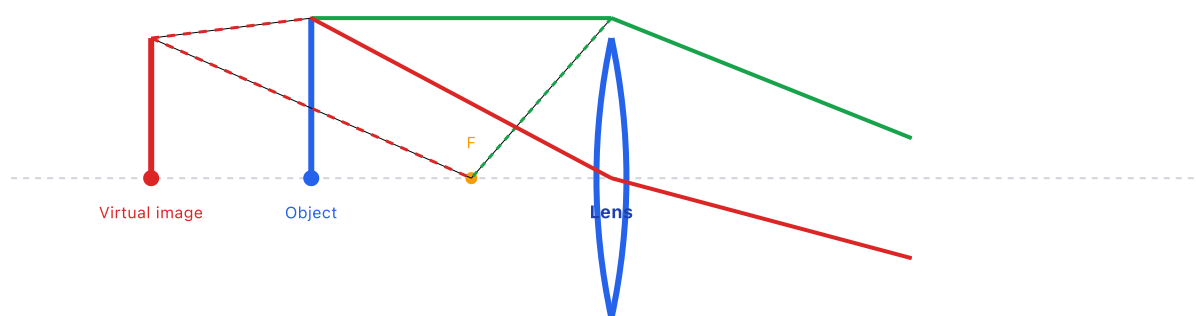


Converging Lens: Virtual Image (object inside f , magnifying glass)



Virtual image is upright, magnified, behind lens. Cannot be projected.

Diverging Lens



Diverging lens always produces virtual, upright, diminished image.

Exam Tip: For converging lenses, remember: object BEYOND $f \rightarrow$ real, inverted image; object INSIDE $f \rightarrow$ virtual, upright, magnified image. For diverging lenses, always virtual, upright, and smaller. Ray diagrams must show all three rays meeting (or appearing to meet for virtual images).

4.6.9 Visible Light and Colour (Triple)

White light is a mixture of all visible colours (wavelengths from ~ 380 nm to ~ 700 nm). It can be split into a spectrum using a prism or diffraction grating. The colours, in order of decreasing wavelength, are: red, orange, yellow, green, blue, indigo, violet (ROYGBIV).

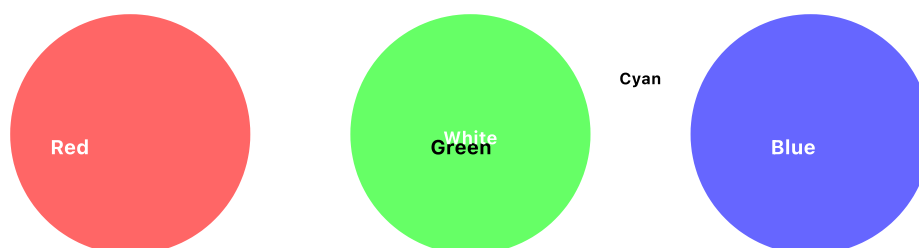
Additive Colour Mixing (Light)

When lights of different colours are mixed, their colours **add**. The primary colours of light are **red (R), green (G), and blue (B)**:

- **Red + Green = Yellow**
- **Green + Blue = Cyan** (light blue)
- **Red + Blue = Magenta** (purple)
- **Red + Green + Blue = White**
- **No light = Black**

This is how computer monitors and TVs create all colours using three colours of pixels.

Additive Colour Mixing (Light)



Opaque Objects and Colour

An **opaque object** reflects certain colours (wavelengths) and absorbs others. The colour we see is the colour it reflects. For example, a red apple appears red because it reflects red light and absorbs blue, green, and other colours.

Examples:

- A **blue book** reflects blue light, absorbs red and yellow
- A **black object** absorbs all colours (no visible light reflected)
- A **white object** reflects all colours roughly equally

Colour Filters

A **colour filter** is a transparent or translucent material that transmits (lets through) certain colours and absorbs others.

- A **red filter** transmits red light, absorbs blue and green
- A **green filter** transmits green light, absorbs red and blue

Colour Under Different Lights

Important principle: An object can only reflect colours that are present in the incident light. If a colour is not in the incoming light, it cannot be reflected, even if the object normally reflects that colour.

Example: A red apple under green light:

- Green light is shining on it
- The apple does not reflect green (it absorbs green in normal white light)
- No other colours are present in the green light
- Result: The apple appears **black** (no visible light is reflected)

Another example: A blue object under red light → appears black (no red to reflect).

Transparency, Translucency, Opacity

- **Transparent:** Light passes straight through; you can see clearly through it (glass, clear water, air)
- **Translucent:** Light passes through but is scattered; you can see light but not clear detail (frosted glass, wax paper)
- **Opaque:** Light does not pass through; no light emerges (wood, metal, brick)

4.6.10 Black Body Radiation (Triple)

A **black body** is an ideal object that absorbs all electromagnetic radiation incident on it (no reflection). A black body is also a perfect emitter — it radiates energy as effectively as any object at the same temperature. All real objects absorb some radiation and reflect some, but many objects approximate black body behaviour.

Emission of Radiation by All Objects

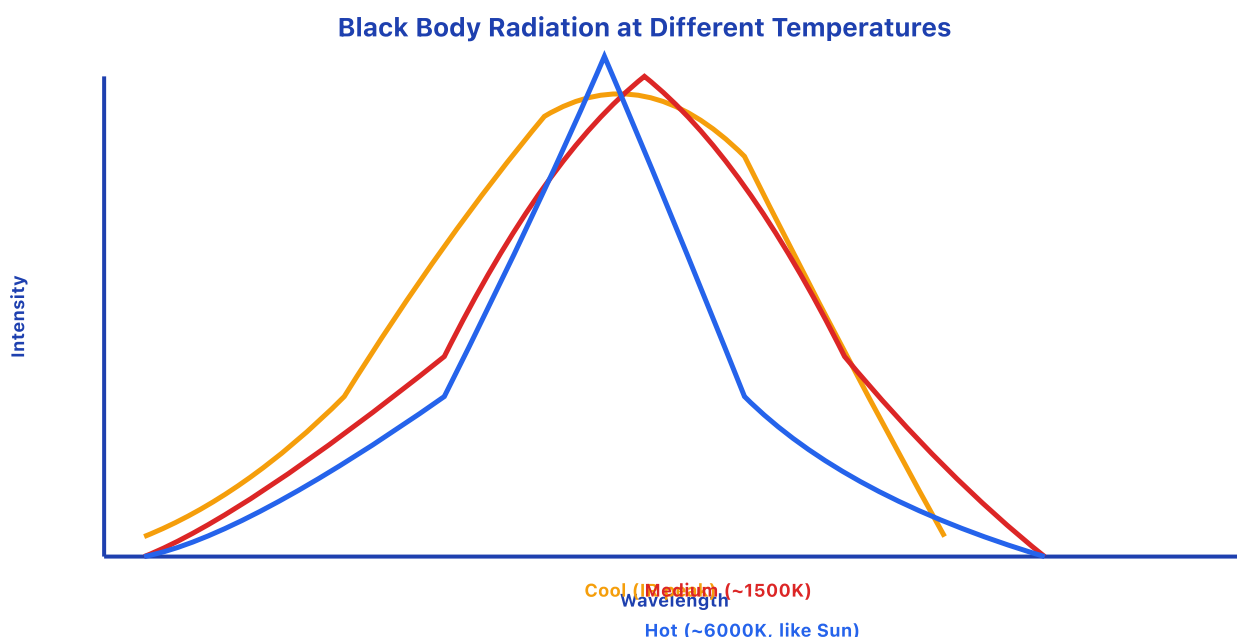
All objects with a temperature above absolute zero (-273°C or 0 K) emit electromagnetic radiation continuously. The dominant wavelength and intensity depend on temperature:

- Cool objects (room temperature, $\sim 20^{\circ}\text{C}$): emit mostly **infrared (heat) radiation** — invisible to the human eye
- Hot objects ($\sim 1000^{\circ}\text{C}$): emit visible light (red/orange glow) plus infrared
- Very hot objects (Sun's surface, $\sim 6000^{\circ}\text{C}$): emit peak intensity in visible white light, plus UV and IR

Wien's Displacement Law (Qualitative)

As an object heats up:

- **Total intensity (area under curve) increases:** The object radiates more total energy
- **Peak wavelength shifts to shorter wavelengths:** The curve "moves left" on the spectrum, from infrared $\rightarrow$ red $\rightarrow$ white $\rightarrow$ blue/UV



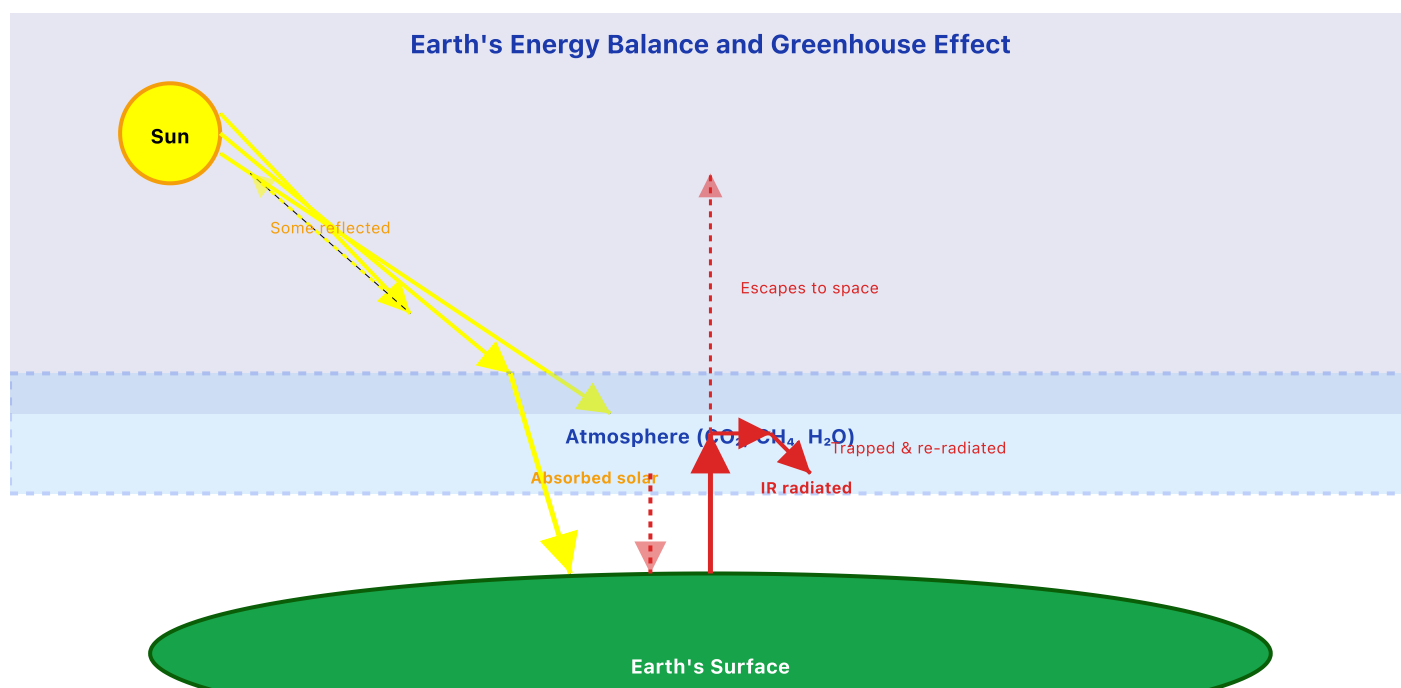
Earth's Energy Balance and the Greenhouse Effect

Solar energy input: The Sun emits radiation across the spectrum. Earth intercepts short-wavelength visible light and ultraviolet radiation. Some is reflected back to space by clouds, ice, and the atmosphere (albedo).

Earth's radiation output: Earth's surface is heated by the absorbed solar energy and radiates energy back outward. Because Earth is much cooler than the Sun ($\sim 15^{\circ}\text{C}$ at surface, $\sim 6000^{\circ}\text{C}$ at Sun's surface), Earth's radiation is long-wavelength **infrared (heat)**.

The Greenhouse Effect: Gases in the atmosphere (CO_2 , CH_4 , H_2O) are relatively transparent to incoming short-wavelength visible light from the Sun but absorb outgoing long-wavelength infrared radiation from Earth. This traps heat in the atmosphere:

1. Solar energy enters and heats Earth
2. Earth radiates infrared back outward
3. Greenhouse gases absorb ~50% of this infrared and re-radiate it downward
4. This warms the atmosphere and surface further (the greenhouse effect)



Exam Tip: Remember: the greenhouse effect is enhanced because humans burn fossil fuels, adding more CO_2 and other greenhouse gases to the atmosphere. This increases the proportion of Earth's infrared radiation that is trapped, leading to global warming. It is NOT a new effect — the natural greenhouse effect has always existed and keeps Earth warm enough for life. The concern is the ENHANCED effect from increased gas concentrations.

Albedo is the fraction (or percentage) of incident solar radiation reflected by a surface. Bright surfaces (ice, snow, clouds) have high albedo (~0.8–0.9); dark surfaces (soil, ocean, forests) have low albedo (~0.1–0.3). Ice melting reduces Earth's albedo, increasing warming (positive feedback).

Summary Table: Wave Types and Properties

Wave Type	Direction of Oscillation	Media	Speed (approx)	Examples
Transverse	Perpendicular to travel	Solids, liquids, vacuum	Varies (light: 3×10^8 m/s)	Water waves, EM waves, string waves
Longitudinal	Parallel to travel	Solids, liquids, gases	Air: ~ 340 m/s; water: ~ 1500 m/s; steel: ~ 5000 m/s	Sound, P-waves, compressions

Intensity is the power per unit area carried by a wave. It is related to amplitude: doubling amplitude quadruples the intensity ($I \propto A^2$). Louder sounds have greater intensity; brighter lights have greater intensity.

Exam Tip: Always check the units in calculations: wavelength in m, frequency in Hz, speed in m/s. For ultrasound and seismic waves, remember to divide the distance by 2 if using round-trip time. For lenses, object distance u and image distance v are measured from the lens centre, along the optical axis.